ABBA4 Implicit: Exterior Projection

One symmetric projection around an unprojected triple jump

1 Unprojected fourth-order base

Let $\mathcal{A}_{q,t}$ be one signed endpoint-time ABBA map on two copies, with no intermediate diagonal projection. Define

$$\gamma = \frac{1}{2 - \sqrt[3]{2}}, \qquad \delta = -\frac{\sqrt[3]{2}}{2 - \sqrt[3]{2}}.$$

The complete base map is

$$\mathcal{C}_{h,t} = \mathcal{A}_{\gamma h, t+(\gamma+\delta)h} \circ \mathcal{A}_{\delta h, t+\gamma h} \circ \mathcal{A}_{\gamma h, t}.$$

Because $2\gamma + \delta = 1$ and $2\gamma^3 + \delta^3 = 0$, the palindromic composition cancels the leading third-order defect of a symmetric second-order base. The middle duration is negative, and the two copies pass continuously through all three factors.

2 One outer symmetric projection

For

$$E = \begin{pmatrix} I \\ I \end{pmatrix}, \quad P = \tfrac{1}{2}\begin{pmatrix} I & I \end{pmatrix}, \quad G = \begin{pmatrix} I & -I \end{pmatrix}, \quad N = G^T,$$

the reduced formulation is

$$M(\mu) = \mathcal{C}_{h,t}(Ez + N\mu), \tag{1}$$
$$r(\mu) = GM(\mu) + 2\mu = 0, \tag{2}$$
$$z^+ = P(M(\mu) + N\mu). \tag{3}$$

If J_j is the Jacobian of signed factor j, then $J_{\mathcal{C}} = J_3 J_2 J_1$ in execution order and

$$D_\mu r = G J_{\mathcal{C}} N + 2I.$$

Every residual evaluation traverses all three ABBA factors.

3 Configurations

The sole implementation of `ABBA4Implicit` exposes reduced or simultaneous spatial projection with a configured `NewtonSolver` or `BroydenSolver`. Only physical coordinates are duplicated. Execution injects the vector field and optional derivative; the method prepares a `DoubleFormulation` and one outer equation. The solver owns tolerances, differentiation policy and iteration controls. In the simultaneous formulation the corrected doubled output joins μ in the unknown; elimination gives the reduced residual above. Optional `EnergyObserver` quadrature owns physical κ separately and uses the accepted signed shear trace. Neither time nor momentum enters a projection root, and neither projection choice introduces intermediate projection.

4 Removed per-map implementation

The current default uses one outer projection. The former `after_each_abba_map` implementation, with three distinct roots and intermediate diagonal re-embedding, has been removed. It defined a different numerical map, since projection and factor evolution do not commute. An explicit request for that placement now raises an error. The legacy single-projection factory is removed. Use `ABBA4Implicit` for the single outer-projection method.

5 Properties and limitations

The ideal method is symmetric and designed for order four. Geometric claims refer to the converged projection root; finite nonlinear tolerances perturb the map. Negative internal time increments require the potential and dynamics to be evaluable slightly outside the outer interval. Observer records preserve the three ordered factors so the accepted-map tangent can be reconstructed.

6 Implementation correspondence

The public `ABBA4Implicit` configuration is in `src/methods/palindromic/abba4_implicit.py`. Its shared advance lives in `src/methods/palindromic/projected_base.py`; the outer equation is in `src/methods/projection.py`, bound to the composition by `src/methods/palindromic/core/projection.py`. Diagnostic observation consumes accepted records through `src/diagnostics/methods/`. The older single-projection import adapter and deprecated factory have been removed. Use `ABBA4Implicit` and its canonical observation record. The canonical architecture document `../simulation/abba4-implicit-simulation-architecture.md` gives the dimensions, diagnostics, and the common prepared integration lifecycle.